Weather Forecasting (RNN/LSTM) & CNN Filter Visualization

Deep Learning I: Neural Networks · Semester 4 · Final Project

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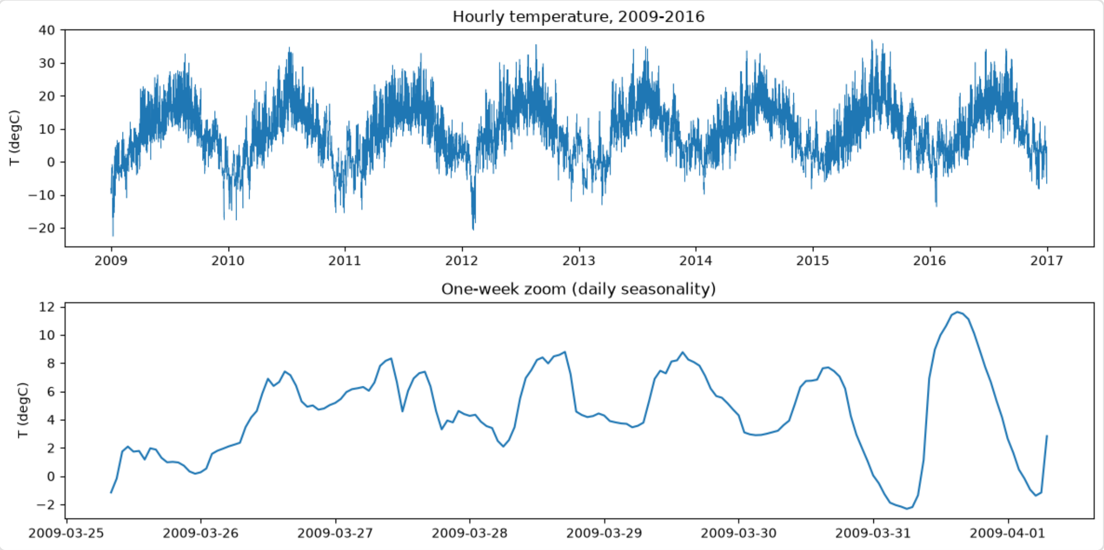
Agenda — Two Tasks

- **Task 1 (Applied):** Forecast temperature 12 h ahead — baseline vs RNN vs LSTM
- **Task 2 (Experimental):** Visualize what a CNN learns on CIFAR-10
- Proper evaluation throughout: loss curves, error metrics, confusion matrix, ablation

TASK 1 · TIME-SERIES

Forecasting Jena Weather, 12 h Ahead

- Jena Climate (2009–2016), 10-min readings → hourly (~70k points)
- Target: air temperature; strong **daily** seasonality



TASK 1 · TIME-SERIES

Setup — No Data Leakage

- **Chronological** 70/15/15 split (never shuffle time → no peeking ahead)
- Standardised using **train-only** mean/std; metrics reported back in °C
- Sliding windows: last **LOOKBACK** hours → value 12 h later

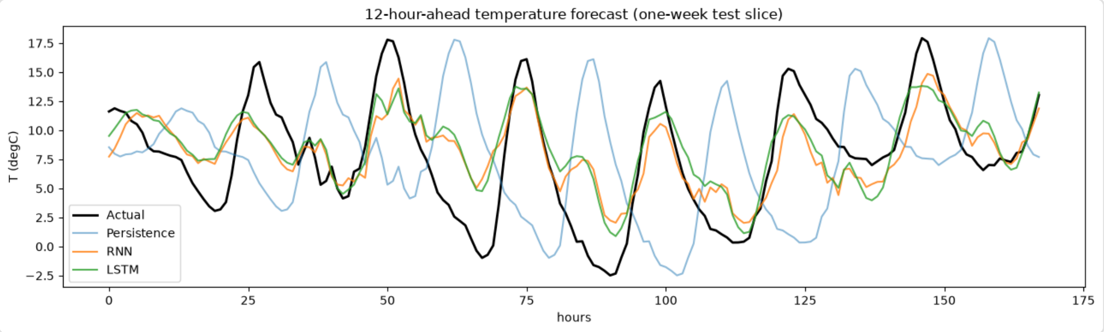
TASK 1 · TIME-SERIES

Three Approaches, Compared Fairly

- **Baseline:** persistence (next = last) & moving average — no training
- **Vanilla RNN** (hidden 64) and **LSTM** (hidden 64)
- Identical except the recurrent cell → clean comparison; Adam + MSE, 12 epochs

TASK 1 · TIME-SERIES

LSTM Wins — Beats Persistence by ~48%

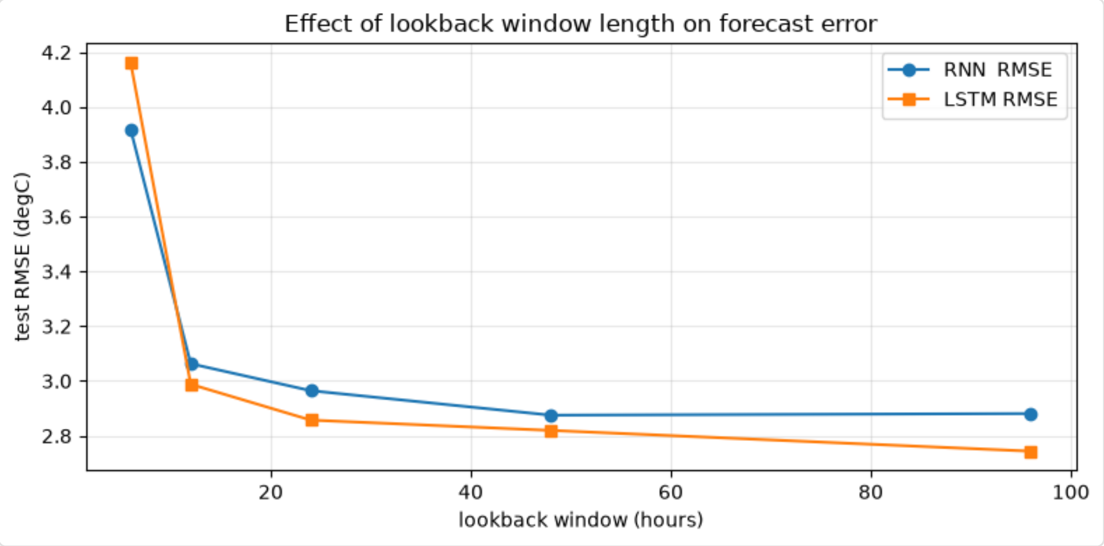


- RMSE (°C): **LSTM 2.81** < RNN 2.86 < moving-avg 3.73 < persistence 5.41
- Persistence is a full half-day **out of phase** at 12 h

TASK 1 · TIME-SERIES

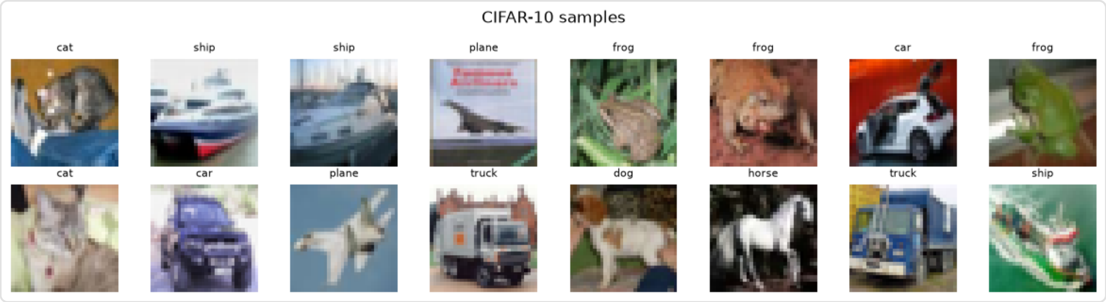
Ablation — Lookback Window Length

- Swept lookback = 6 / 12 / 24 / 48 / 96 h
- 6 h too short → error drops sharply by ~24 h → flattens
- **LSTM keeps improving to 96 h**; RNN plateaus



TASK 2 · CNN VISUALIZATION

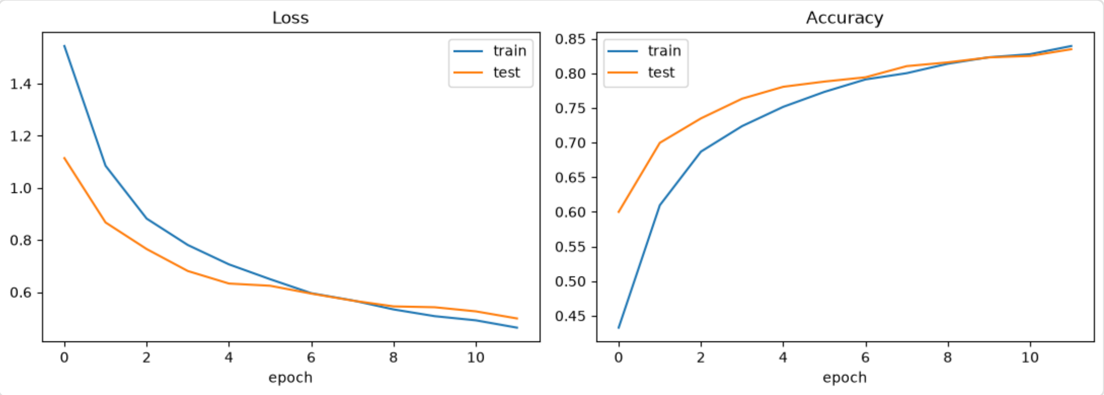
A Small VGG-style CNN on CIFAR-10



- 10 classes, 32×32 RGB; 4 conv layers (32→64→128→128) + FC head; ~2.3M params
- Light augmentation (crop + flip) + channel normalisation

TASK 2 · CNN VISUALIZATION

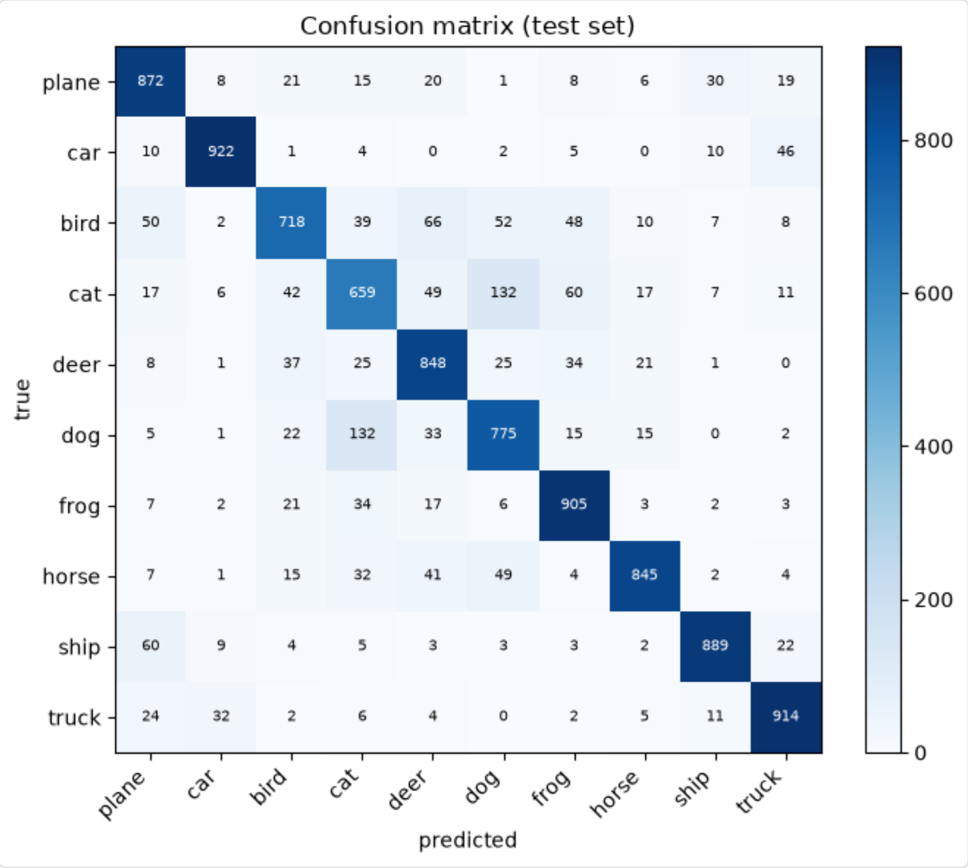
~82% Test Accuracy in 12 Epochs



- Healthy curves, small train/test gap (not overfitting)

Confusion Matrix — Where It Errs

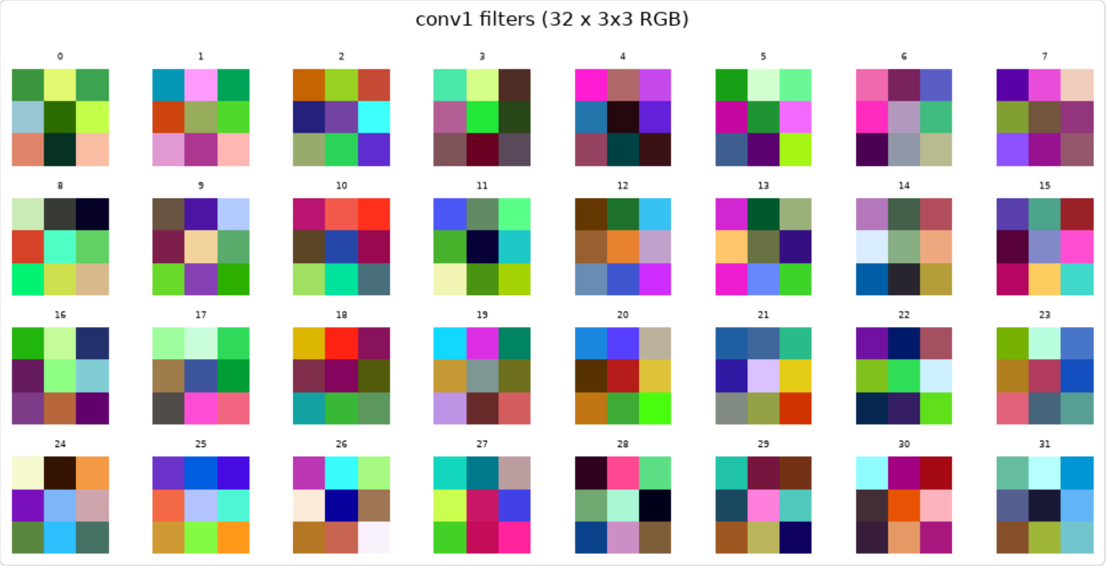
- Errors cluster between similar classes: **cat↔dog, car↔truck**
- Hardest: cat 61%, bird 70% · Easiest: ship 92%, plane 90%



TASK 2 · CNN VISUALIZATION

First-Layer Filters (conv1)

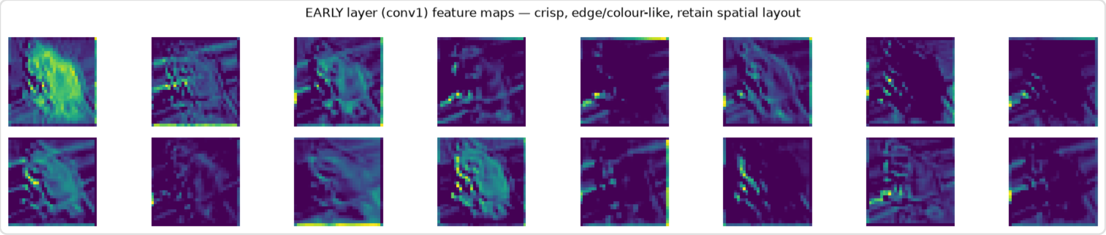
- 32 filters (3×3 RGB) → **oriented edges, colour-opponent blobs**
- Low-level detectors operating directly on raw pixels



Feature Maps — Early vs Late

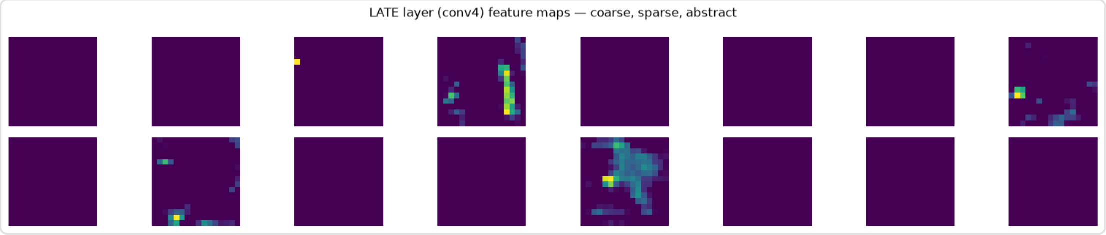
Early (conv1)

sharp, retains outline → *where*



Late (conv4)

coarse, sparse, abstract → *what*



Why Depth = Abstraction

- Hierarchy: pixels → edges/colours → textures/parts → class concepts
- Growing receptive field + pooling trade spatial detail for semantic meaning
- This is **why** deep conv stacks classify well (and why cat↔dog still confuse)

Conclusions & Q&A

- TS: LSTM > RNN ≫ baselines at 12 h; lookback sweet-spot ≈ daily cycle
- CNN: 82% accuracy; visualizations show clear depth→abstraction
- Future: multi-step + exogenous features (TS); deeper net / Grad-CAM (CNN)
- **Thank you — Questions?**